Bayesian Poisson Regression for Defect Prediction from Storage Time: Inference, Testing and Risk Bands

Olivier P.

Abstract

With about 100 products, we model the number of defects Y as a function of storage time X by a Bayesian Poisson regression with log link. We prove that the posterior is strictly log-concave, which justifies Laplace and Hamiltonian Monte Carlo inference; we show that the naive quantity $P(\beta_1 \neq 0 \mid \mathcal{D})$ is identically 1 under a continuous prior and replace it by directional probabilities, a region of practical equivalence and a Savage–Dickey Bayes factor; we derive Rao–Blackwellised estimators of the probability that a product falls into a defect band, prove that these probabilities are monotone in storage time, and turn them into a maximal storage time at a given risk level. Conditions for a causal reading and model checks (overdispersion, WAIC, LOO) are made explicit.

1. Model

Data are $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, $x_i \geq 0$ days, $y_i \in \mathbb{N}$ defects, assumed conditionally independent given the parameters. A generalised linear model [6] with canonical log link reads

$$Y_i \mid x_i, \beta \sim \mathrm{Pois}(\lambda_i), \qquad \eta_i = \log \lambda_i = \beta_0 + \beta_1 x_i, \quad (1)$$

so e^{β_1} is the multiplicative change in the defect rate per day of storage, and $e^{30\beta_1}$ per month. The prior is $\beta \sim \mathcal{N}(0, \Sigma_0)$, $\Sigma_0 = \mathrm{diag}(\sigma_0^2, \sigma_1^2)$, on standardised x so that σ_1 has a unit-free meaning (e.g. $\sigma_1 = 1$: a one-standard-deviation increase in storage time is a priori unlikely to multiply the rate by more than e^2).

2. Posterior geometry

With $\tilde{x}_i = (1, x_i)^\top$, Bayes' theorem gives

$$\log p(\beta \mid \mathcal{D}) = \sum_i \big(y_i \eta_i - e^{\eta_i}\big) - \tfrac{1}{2}\beta^\top \Sigma_0^{-1} \beta + \text{const.} \quad (2)$$

Proposition 1. *The posterior is strictly log-concave: its Hessian is $-\big(\sum_i e^{\eta_i} \tilde{x}_i \tilde{x}_i^\top + \Sigma_0^{-1}\big) \prec 0$. Hence it is unimodal, the MAP $\hat{\beta}$ is unique, and Newton's method converges to it.*

Proof. $\partial^2_\beta (y_i \eta_i - e^{\eta_i}) = -e^{\eta_i} \tilde{x}_i \tilde{x}_i^\top \preceq 0$ and $\Sigma_0^{-1} \succ 0$. □

The Laplace approximation is $\beta \mid \mathcal{D} \approx \mathcal{N}(\hat{\beta}, V)$, $V = (\tilde{X}^\top \hat{W} \tilde{X} + \Sigma_0^{-1})^{-1}$, $\hat{W} = \mathrm{diag}(\hat{\lambda}_i)$; by Bernstein–von Mises [9] it is asymptotically exact. For $n \approx 100$ we use exact sampling by Hamiltonian Monte Carlo [7] with the No-U-Turn sampler [3], whose efficiency on log-concave targets is well understood; convergence is checked with $\hat{R}$ and effective sample sizes [2].

Design. From V, $\mathrm{Var}(\beta_1 \mid \mathcal{D}) \approx \big(\sum_i \hat{\lambda}_i (x_i - \bar{x}_\lambda)^2\big)^{-1}$, with $\bar{x}_\lambda$ the $\hat{\lambda}$-weighted mean of x: precision grows with the spread of storage times. Inspecting products with very different storage times is the cheapest way to learn β_1.

3. Is there a link?

A degenerate quantity. Under any prior with a density, $P(\beta_1 = 0) = 0$, hence $P(\beta_1 = 0 \mid \mathcal{D}) = 0$ and $P(\beta_1 \neq 0 \mid \mathcal{D}) = 1$ for every data set; a Monte Carlo estimate with a threshold $|\beta_1| > 10^{-6}$ returns 1 up to sampling noise and is uninformative. We use instead:

(a) Direction: $P(\beta_1 > 0 \mid \mathcal{D}) \approx \frac{1}{S}\sum_s \mathbf{1}\{\beta_1^{(s)} > 0\}$ from S posterior draws.

(b) Practical equivalence [5]: fix ε such that $|\beta_1| < \varepsilon$ is negligible for the plant (e.g. less than 1% rate change per month: $\varepsilon = \ln(1.01)/30$ on the raw scale) and report $P(|\beta_1| < \varepsilon \mid \mathcal{D})$.

(c) Bayes factor [4] between $M_0 : \beta_1 = 0$ and M_1. If the prior of β_0 under M_0 equals the conditional prior of β_0 given $\beta_1 = 0$ under M_1 (true here, by independence), the Savage–Dickey identity [1] gives

$$\mathrm{BF}_{01} = \frac{p(\mathcal{D} \mid M_0)}{p(\mathcal{D} \mid M_1)} = \frac{p(\beta_1 = 0 \mid \mathcal{D}, M_1)}{p(\beta_1 = 0 \mid M_1)}, \quad (3)$$

the ratio of posterior to prior density at 0, estimated by kernel density on the draws or by the Laplace approximation. Unlike (a), it is sensitive to σ_1, which must be justified.

Causality. β_1 is an association. It is the causal effect of storage time if storage time is ignorable given covariates Z, i.e. no common cause of X and Y (product family, supplier, season) is left out; by the back-door criterion [8] one then fits $\eta_i = \beta_0 + \beta_1 x_i + \gamma^\top z_i$. A randomised storage experiment removes the need for this assumption.

4. Prediction and risk bands

The posterior predictive law of a new product stored $\tilde{x}$ days is $p(\tilde{y} \mid \tilde{x}, \mathcal{D}) = \int \mathrm{Pois}(\tilde{y}; e^{\beta_0 + \beta_1 \tilde{x}})\, p(\beta \mid \mathcal{D})\, d\beta$. For a band $B = \{a, \dots, b-1\}$ (e.g. $Y < 5$, $5 \leq Y < 10$, $Y \geq 10$) we use the Rao–Blackwellised estimator

$$\hat{P}(\tilde{Y} \in B \mid \tilde{x}, \mathcal{D}) = \frac{1}{S}\sum_{s=1}^{S} \sum_{y \in B} \mathrm{Pois}\big(y; \lambda^{(s)}(\tilde{x})\big), \quad (4)$$

which is unbiased for the posterior predictive probability and, by the law of total variance, has variance no larger than that of the estimator based on simulated $\tilde{y}^{(s)}$; its Monte Carlo error is $O(S_{\mathrm{eff}}^{-1/2})$.

Proposition 2 (monotone risk)**.** *If $P(\beta_1 > 0 \mid \mathcal{D}) = 1$, then $\tilde{x} \mapsto P(\tilde{Y} \geq k \mid \tilde{x}, \mathcal{D})$ is non-decreasing for every k.*

Proof. $\lambda \mapsto P(\mathrm{Pois}(\lambda) \geq k)$ is non-decreasing (its derivative is $\mathrm{Pois}(k-1; \lambda) \geq 0$), and $\tilde{x} \mapsto \lambda(\tilde{x})$ is non-decreasing when $\beta_1 > 0$; integrate over the posterior. □

When $P(\beta_1 > 0 \mid \mathcal{D})$ is close to but below 1 the property holds up to a small correction. It yields a decision rule: the maximal storage time at risk level α,

$$x^*(\alpha) = \sup\{x : \; P(\tilde{Y} \geq 10 \mid x, \mathcal{D}) \leq \alpha\}, \quad (5)$$

which accounts for parameter uncertainty, unlike a plug-in rule $\lambda(x;\hat\beta)$. If costs c_d per defect and c_s per day are known, the Bayes action minimises $c_d\,\mathbb{E}[\tilde Y \mid x,\mathcal{D}] - c_s\,x$ instead.

5. Model checking

Overdispersion. The Poisson model imposes $\mathrm{Var}(Y \mid x) = \mathbb{E}(Y \mid x)$. We compare the observed dispersion statistic $\sum_i (y_i-\lambda_i)^2/\lambda_i$ with its posterior predictive distribution; if it is too large, we use the Gamma–Poisson (negative binomial) model, $\mathrm{Var} = \lambda + \lambda^2/\kappa$ [2].

Comparison. Alternative predictors ($\eta = \beta_0 + \beta_1 x + \beta_2 x^2$, covariates, negative binomial) are compared by WAIC [11], $\mathrm{WAIC} = -2\sum_i \log \frac{1}{S}\sum_s p(y_i \mid \beta^{(s)}) + 2\sum_i \mathrm{Var}_s \log p(y_i \mid \beta^{(s)})$, and by Pareto-smoothed leave-one-out cross-validation [10], whose diagnostics also flag influential products. **Prior sensitivity** is assessed by refitting with $\sigma_1 \in \{0.5, 1, 10\}$: (a) and (b) should be stable, (c) will not be.

6. Status

Model, inference, tests and decision rule are specified, with pseudo-code for posterior sampling and predictive bands; next is the implementation on plant data, with covariates and overdispersion checks.

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